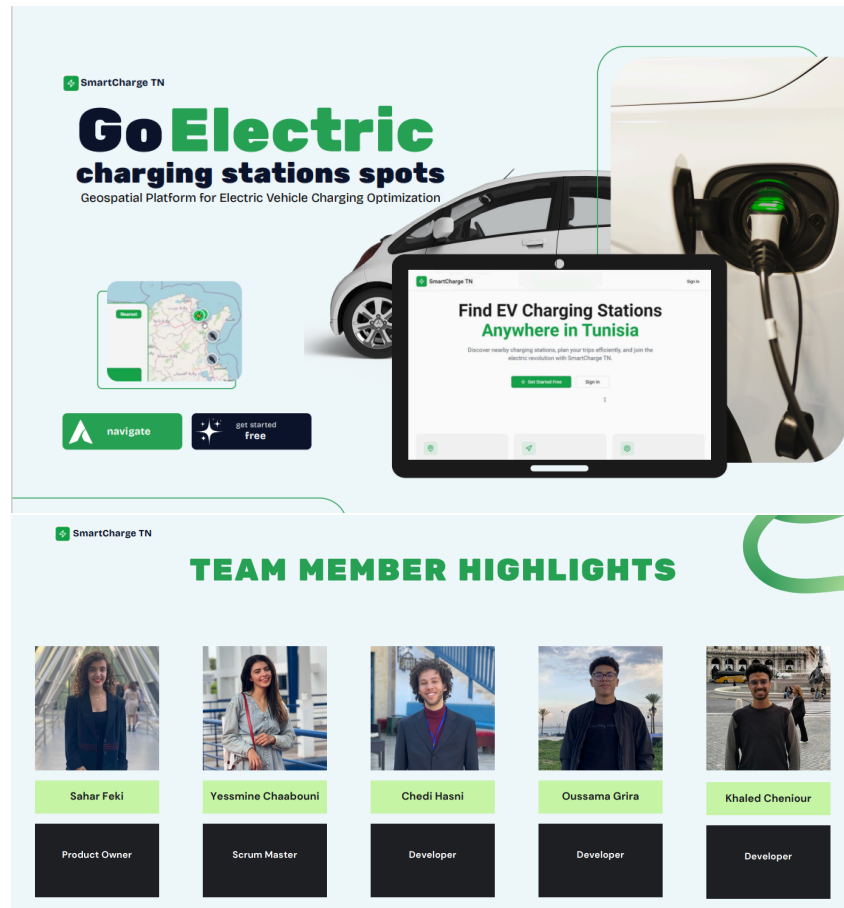


SmartCharge TN

Geospatial Platform for Electric Vehicle Charging



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1 Context and Problem Statement

1.1 Project Overview

SmartCharge TN is a real-time geospatial platform designed to optimize Electric Vehicle (EV) charging accessibility across Tunisia. It enables EV drivers to discover nearby charging stations, plan efficient routes, and join Tunisia's green mobility revolution.

1.2 Motivation

The number of EV users in Tunisia has increased significantly over recent years, yet the lack of real-time, centralized information about charging stations remains a key barrier. This project addresses:

- Difficulty locating available chargers in real-time
- Absence of a unified Tunisian EV charging database
- Range anxiety during intercity travel
- Station concentration in major cities leaves intercity travel underserved

1.3 How EV Owners Act Today

- Search manually on Google Maps (limited coverage)
- Depend on foreign apps like PlugShare, ChargeMap (lacks local data)
- Rely on Facebook groups & random word-of-mouth
- Use STEG Website (static list, not interactive)

1.4 Challenge of Electric Mobility in Tunisia

- No real-time, centralized database for charging stations
- Drivers experience "range anxiety" without live data or optimized routes
- No unified system showing availability, type, and status

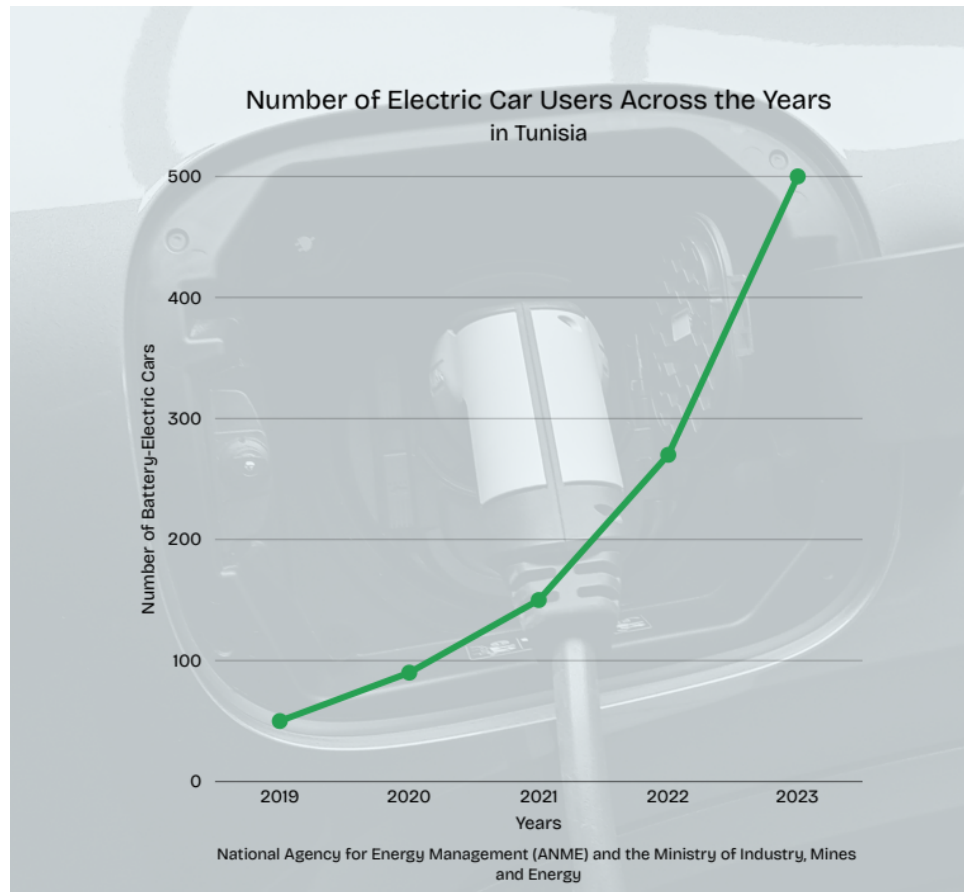


Figure 1.1: Number of Electric Car Users in Tunisia (2019-2023)

2 Project Presentation

2.1 Our Solution

SmartCharge TN provides:

- Real-time interactive EV charging map showing accurate charger availability, type, and status
- Route optimization with smart navigation
- Filters for charger type, network, and availability
- Web and mobile application (Android/iOS)
- Data dashboards for energy operators on charging demand and load distribution

2.2 Value Proposition

- **For EV Drivers:** Real-time map with route optimization
- **For Energy Operators:** Data dashboards on charging demand, load distribution, and maintenance
- **For Government/Municipalities:** Insights for urban planning and CO reduction tracking

2.3 Business Model Canvas

2.3.1 Mission – Vision – Raison d’être – Ambition

- **Mission:** Connect EV drivers to the nearest, available charging stations in Tunisia via a real-time platform.

- **Vision:** The leading smart mobility platform and electric transport reference in North Africa by 2030.
- **Raison d'être:** Drive Tunisia's green transition by eliminating EV "charging uncertainty."
- **Ambition:** Integrate with Tunisia's national digital mobility network and expand regionally through partnerships.

2.3.2 Stakeholders

- EV owners (Tesla, BYD, Renault Zoe users)
- STEG and private energy operators
- Municipalities (Tunis, Sfax, Sousse)
- Ministry of Environment, Ministry of Transport, Ministry of ICT
- Startups in renewable energy ecosystem (e.g., WattNow, Touensa Green)

2.3.3 Revenue Model

- **Freemium model:** Free basic use, premium paid route optimization (=10 TND/month)
- **B2B subscriptions:** Analytics dashboards (=500–1000 TND/month per client)
- **Advertising & sponsorships** from EV brands or green tech
- **Data insights** to energy providers and municipalities

2.3.4 Revenue Redistribution

- 40% reinvested in infrastructure and app maintenance
- 30% to R&D and innovation
- 20% for staff salaries and training
- 10% to public awareness and sustainability campaigns

BUSINESS MODEL CANVAS (BMC)

MISSION - VISION - RAISON D'ÊTRE - AMBITION

Mission: Connect EV drivers to the nearest, available charging stations in Tunisia via a real-time platform.

Vision: The leading smart mobility platform and electric transport reference in North Africa by 2030.

Raison d'être: Drive Tunisia's green transition by eliminating EV "charging uncertainty."

Ambition: Integrate with Tunisia's national digital mobility network and expand regionally through partnerships.

PROBLEM

- No real-time, centralized database for charging stations.
- Drivers experience "range anxiety" without live data or optimized routes.
- Station concentration in major cities leaves intercity travel underserved.

VALUE PROPOSITION

- **EV Drivers:** Real-time map showing accurate charger availability, type, and status with route optimization.
- **Energy Operators:** Data dashboards on charging demand, load distribution, and maintenance.
- **Government/ Municipalities:** Insights for urban planning and CO₂ reduction tracking.

SOLUTION

A web and mobile app providing a real-time, interactive map of charging stations with smart navigation and user filters.

EXISTING ALTERNATIVES

- **Google Maps:** Lists few charging points, no live updates, limited Tunisian coverage.
- **PlugShare / ChargeMap:** Useful abroad but lacks local data and Arabic/French localization.
- **STEG Website:** Shows fixed station list, not interactive.
- **Conclusion:** SmartCharge TN is the first real-time, Tunisia-specific geospatial EV charging platform.

STAKEHOLDERS

- **EV owners** (Tesla, BYD, Renault Zoe users).
- **STEG** and private **energy operators**.
- **Municipalities** (Tunis, Sfax, Sousse).
- **Ministry** of Environment, Ministry of Transport, Ministry of ICT.
- **Startups** in the renewable energy ecosystem (e.g., WattNow, Touensa Green).

CHANNELS

- **Mobile app** (Android/iOS) – main user entry.
- **Web platform** – for municipalities and partners.
- **Social media** campaigns – Instagram, LinkedIn, Facebook.
- **Partnerships** with **car dealerships** (Renault, Hyundai Electric Tunisia).
- **Integration** with **smart city dashboards** (e.g., "Smart Sfax Initiative").

POSITIVE SOCIETAL IMPACTS (DIRECT & INDIRECT)

Direct:

- Reduces CO₂ emissions by facilitating EV adoption.
- Builds public confidence in clean energy.
- Saves users time and energy with efficient routing.

Indirect:

- Stimulates private investment in charging infrastructure.
- Creates green jobs in mobility, GIS, and IoT.
- Supports Tunisia's 2030 National Energy Transition Strategy.

NEGATIVE SOCIETAL IMPACTS (DIRECT & INDIRECT)

Direct :

- Grid strain from increased electricity demand.
- Potential misuse of user location data.

Indirect :

- Digital exclusion of users without smartphones or internet.
- Environmental impact of cloud servers.

Figure 2.1: Business Model Canvas 1.0



Figure 2.2: Business Model Canvas 1.1

3 SCRUM Project Management

3.1 Team Members & Roles

- **Product Owner:** Sahar Feki
- **Scrum Master:** Yessmine Chaabouni
- **Developers:** Chedi Hasni, Oussama Grira, Khaled Cheniour

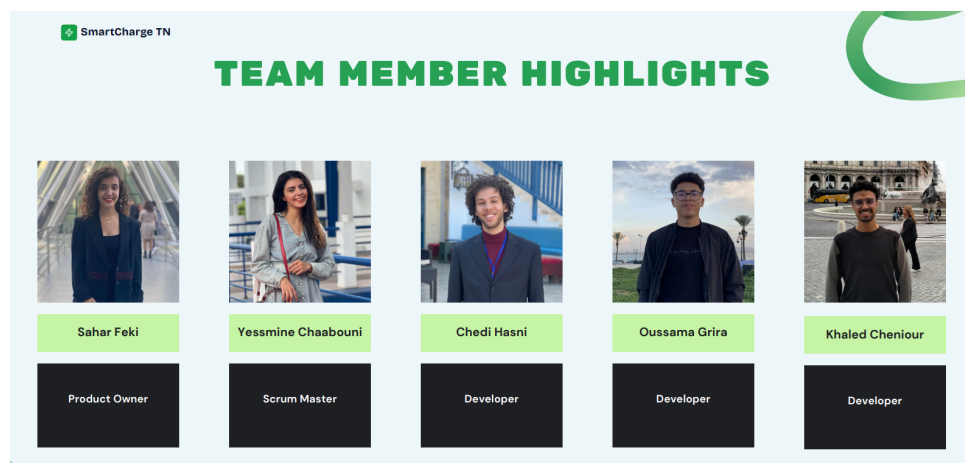


Figure 3.1: SmartCharge TN Team Members

3.2 Use Case Diagram

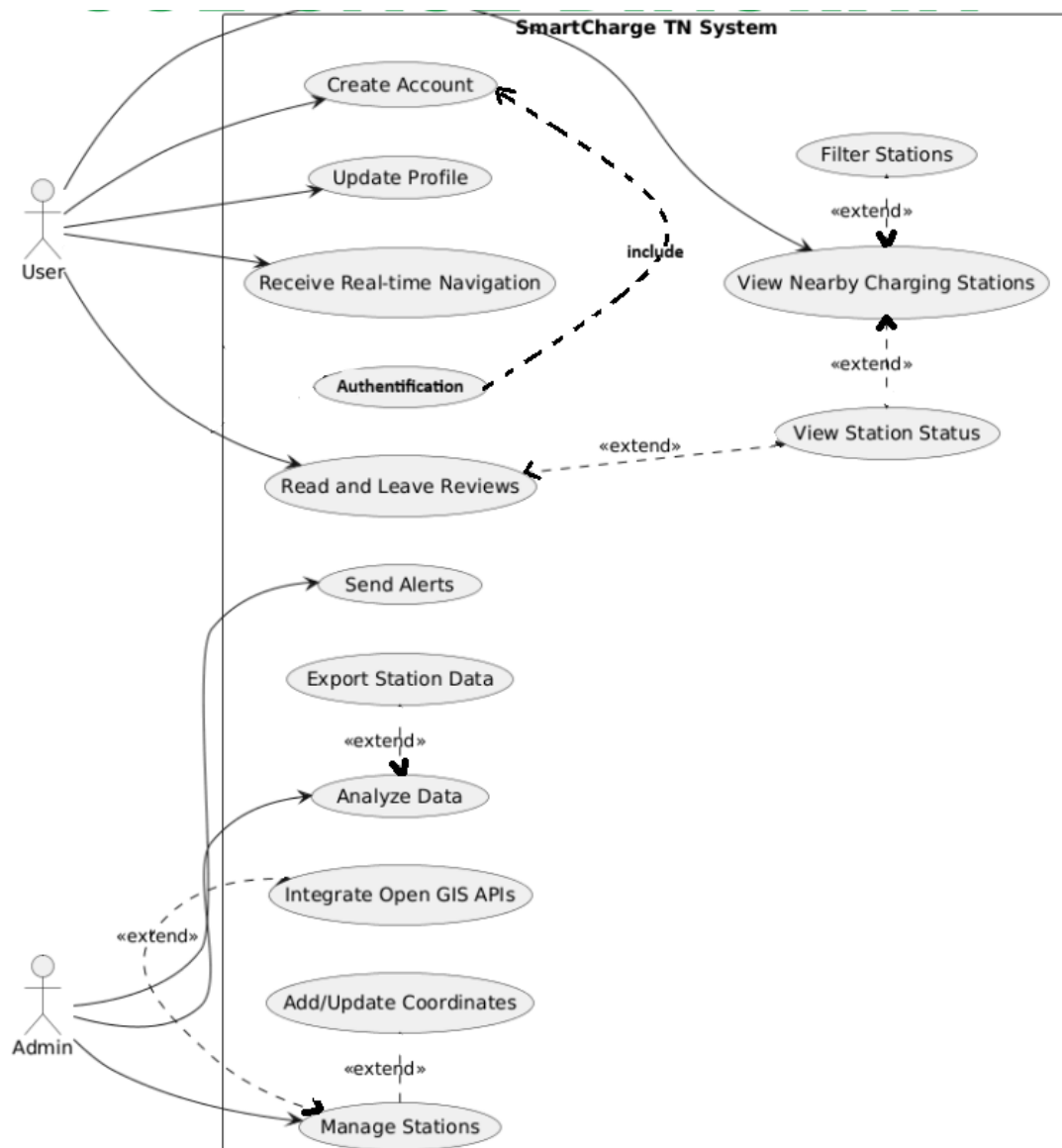


Figure 3.2: SmartCharge TN Use Case Diagram

3.3 Product Backlog

User Story N	As a	I want to	So that	Status	Priority	Est.
1	Charger User	see all nearby charging stations on the map	I can locate the closest charging point	Completed	High	8
2	Charger User	create an account using email or social media	I can access personalized features securely	Completed	High	8
3	Charger User	see the station's real-time status (occupied/free)	I can avoid unavailable chargers	In progress	High	13
4	Admin	integrate the app with open GIS APIs	I can display accurate base maps	Completed	High	8
5	Admin	manage (add/edit/delete) charging stations	I can keep platform data updated	Completed	High	5
6	Admin	add/update station coordinates using geomatics interface	data stays accurate	Completed	High	13
7	Charger User	receive real-time navigation to selected station	I can reach the station efficiently	In progress	High	13
8	Charger User	filter stations by charging type or network	I can find suitable stations for my car	Completed	Medium	5

User Story N	As a	I want to	So that	Status	Priority	Est.
9	Charger User	update my profile (name, vehicle type, battery range)	route suggestions are accurate	Completed	Medium	5
10	Admin	send broadcast alerts (maintenance, new stations)	users stay informed about updates	Not started	Medium	13
11	Admin	analyze usage data (popular stations, routes)	I can make data-driven decisions	Under review	Medium	8
12	User	read and leave reviews about station quality	I can share my experience and help others	In progress	Low	8
13	Admin	export station and usage data for reporting	I can share insights with partners	In progress	Low	5
14	Admin (optional)	integrate payment or booking systems	users can pay or reserve charging slots	Not started	Low	13

3.4 Task Board Table

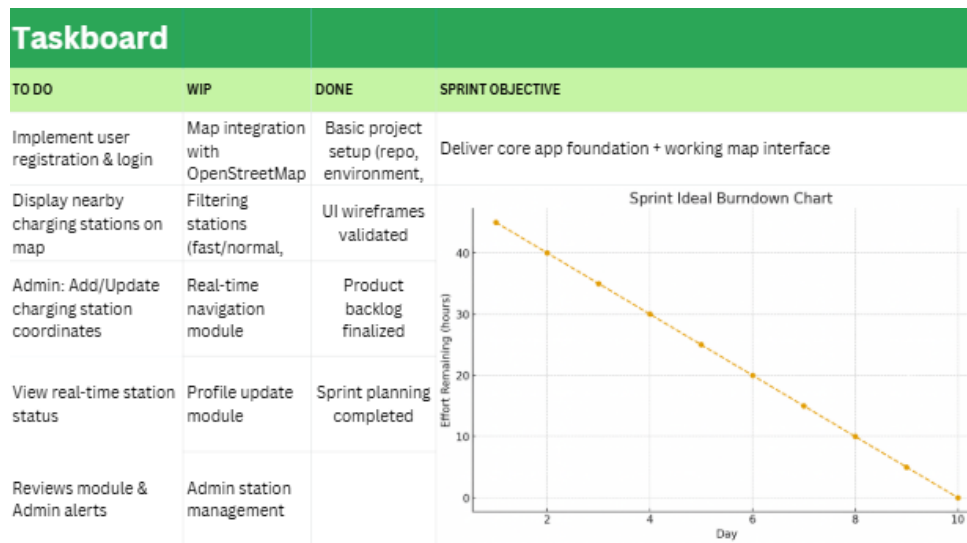


Figure 3.3: Task board Table

3.5 Sprint Plan

3.5.1 Sprint Overview

- **Sprint 1:** Core system setup, map display, admin management (Total Effort: 29)
- **Sprint 2:** Smart navigation, real-time updates, personalization (Total Effort: 49)
- **Sprint 3:** Analytics, user interaction (reviews & export) (Total Effort: 21)
- **Sprint 4:** Alerts, payments (Total Effort: 26)

Sprint Plan			
Sprints			
Sprint 1 (2 weeks)	User Story 2	Sprint 3 (2 weeks)	User Story 12
	User Story 1		User Story 11
	User Story 5		User Story 13
	User Story 4		User Story 10
Sprint 2 (2 weeks)	User Story 9	Sprint 4 (2 weeks)	User Story 14
	User Story 8		
	User Story 7		
	User Story 3		
	User Story 6		

Figure 3.4: Sprint plan Table

3.6 Release Plan

3.6.1 Release 1: Core Platform Launch

Duration: Sprint 1 - 2 (2 weeks)

Objective: Deliver a functional MVP so users can view charging stations and navigate the map.

Included Features:

- User authentication
- Basic user profile (vehicle type, battery range)
- Map integration using OpenStreetMap/Google Maps
- Admin panel: Add/Edit/Delete charging stations
- Basic filtering (fast/normal chargers)
- Static station details (name, type, coordinates)
- Real-time navigation with GPS tracking
- Real-time station status (occupied/free)

Value Delivered: Users can finally find EV chargers in Tunisia with functional map + station list.

Release Plan 1				
Release	Duration	Objective	Included Features	Value Delivered
Release 1: Core Platform Launch	Sprint 1 (2 weeks) 3 Nov - 17 Nov	User Story 2	- User authentication - Basic user profile (vehicle type, battery range) - Map integration using OpenStreetMap/Google Maps - Admin panel: Add/Edit/Delete charging stations - Basic filtering (fast/normal chargers) - Static station details (name, type, coordinates)	- Users can finally find EV chargers in Tunisia - Functional map + station list - Admins can maintain accurate station data - System deployed for internal testing
		User Story 1		
		User Story 5		
		User Story 4		
	Sprint 2 (2 weeks) 3 Nov - 17 Nov	User Story 9	- Real-time navigation - GPS tracking - Filters (type, speed) - Real-time station status - Update profile	- Navigation route visible & updates - Filters dynamically update markers - Status color-coded and refreshed - Profile updates stored
		User Story 8		
		User Story 7		
		User Story 3		
		User Story 6		

Figure 3.5: Release plan 1 Table



Figure 3.6: Graphical View of Interfaces 1.0

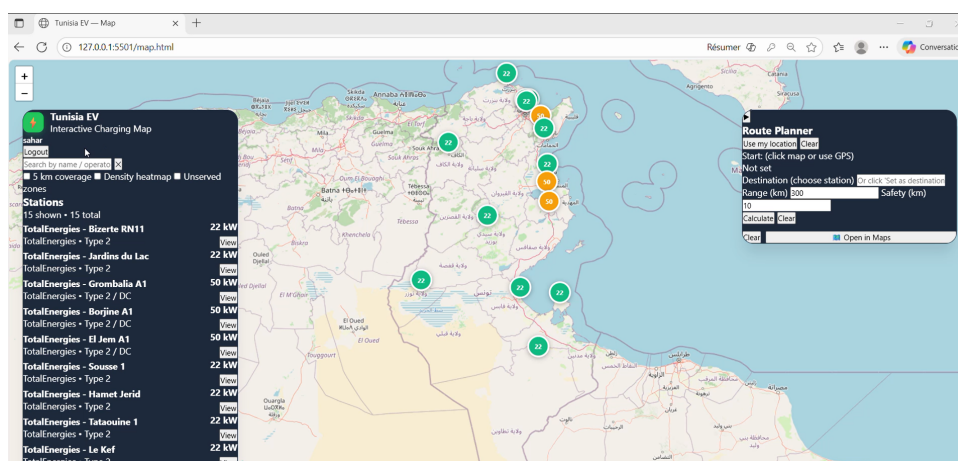


Figure 3.7: Graphical View of Interfaces 1.1

3.6.2 Release 2: Analytics, Engagement & Advanced Features

Duration: Sprint 3 & 4 (4 weeks total)

Objective: Introduce user engagement & analytics, integrate advanced extensions.

Included Features:

- Reviews + Ratings system
- Usage Analytics Dashboard
- Export CSV/PDF functionality
- Admin Alerts system
- Payment/Booking system simulation

Value Delivered: Enhanced user engagement and advanced administrative capabilities.

Release Plan 2					
Release	Duration	Objective		Included Features	Value Delivered
Release 2: Analytics, Engagement & Advanced Features	Sprint 3 (2 weeks) 17 Nov - 1 Dec	User Story 12	Introduce user engagement & analytics	• Reviews + Ratings • Usage Analytics Dashboard • Export CSV/PDF	• Reviews appear under stations • Dashboard visualizes usage • Export formats working
		User Story 11			
		User Story 13			
	Sprint 4 (2 weeks) 17 Nov - 1 Dec	User Story 10	Integrate advanced extensions	• Admin Alerts • Payment/Booking system	• Alerts visible to all users • Payment simulation runs
		User Story 14			

Figure 3.8: Release plan 2 Table

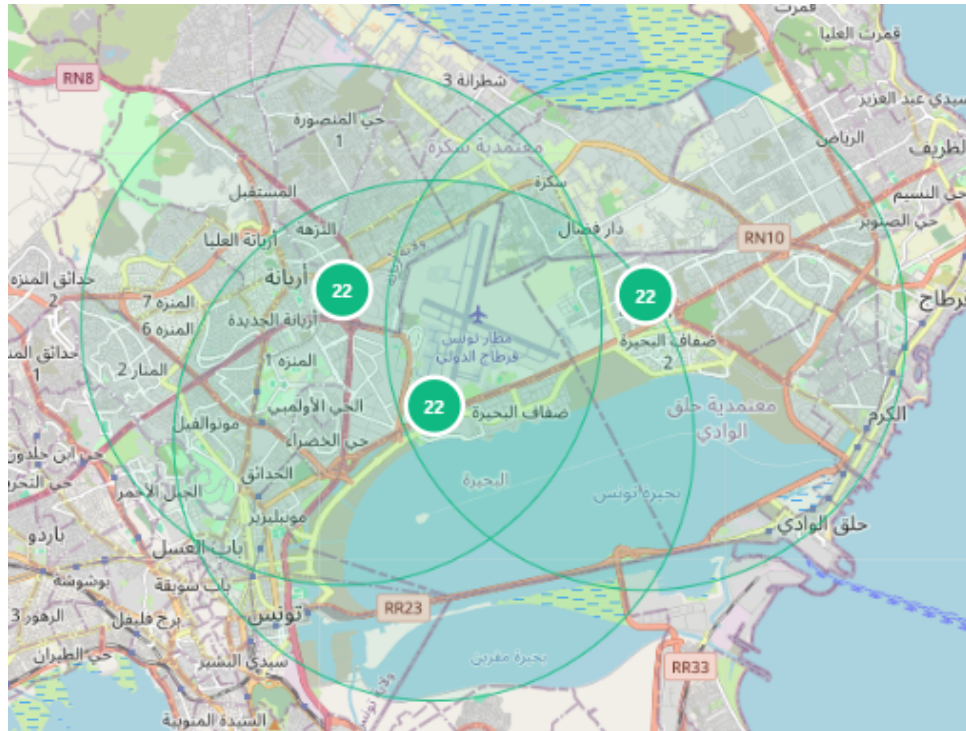


Figure 3.9: Graphical View of Interfaces

4 Sprint Backlogs

4.1 Sprint 1 Backlog - Core Features & Map Display

User Story	Tasks / Sub-tasks	Responsible	Effort	Days Remaining								
User Story 2	• Build registration & login UI • Integrate Firebase/Auth API • Implement email verification • Test login flow	Frontend Dev	8	7	7	6	4	2	1	0		

User Story	Tasks / Sub-tasks	Responsible	Effort	Days Remaining									
User Story 1	• Connect to map API (OSM/Google) • Display markers with names/types • Add zoom & pan • Test map responsiveness	GIS Dev	8	8	8	8	8	8	7	4	3	2	
User Story 5	• Create CRUD interface • Sync with database • Add confirmation alerts • Test data persistence	Backend Dev	5	5	5	5	5	5	5	0			

User Story	Tasks / Sub-tasks	Responsible	Effort	Days Remaining									
User Story 4	• Connect map API keys • Fetch base tiles • Optimize performance on mobile • Test multi-device rendering	GIS Dev	8	7	5	4	3	3	3	3	3	3	2
0													

4.2 Sprint 2 Backlog - Smart Navigation & Personalization

User Story	Tasks / Sub-tasks	Responsible	Effort	Days Remaining									
User Story 9	• Create profile update form • Link to database • Use vehicle data in route logic • Test persistence	Frontend Dev	5	5	4	4	4	4	0				

User Story	Tasks / Sub-tasks	Responsible	Effort	Days Remaining									
User Story 8	• Add filter dropdowns • Connect to station attributes • Update map dynamically • Validate results	GIS Dev	5	4	4	4	4	3	2	1	0		
User Story 7	• Integrate GPS tracking • Draw route line dynamically • Display ETA and distance • Test live updates	Mobile Dev	13	12	4	4	4	6	6	6	6	6	
User Story 3	• Connect to live API • Refresh station data every few seconds • Add color status indicator • Test reliability	Backend Dev	13	10	4	4	4	4	4	4	4	4	

User Story	Tasks / Sub-tasks	Responsible	Effort	Days Remaining									
User Story 6	• Build coordinate entry form • Integrate map picker • Sync with DB • Test update accuracy	GIS Dev	13	10	4	4	4	4	3	2	1	0	

4.3 Sprint 3 Backlog - Data Analysis, Reviews & Export

User Story	Tasks / Sub-tasks	Responsible	Effort	Days Remaining									
User Story 12	• Add rating & comment fields • Connect to DB • Display reviews under station details • Moderate spam inputs	Frontend Dev	8	8	8	8	8	8	8	8	8	8	8

User Story	Tasks / Sub-tasks	Responsible	Effort	Days Remaining									
User Story 11	• Collect usage logs • Create dashboard charts (stations/routes) • Connect analytics DB • Test metrics accuracy	Data Analyst	8	8	7	7	7	7	7	7	7	6	6
User Story 13	• Add "Export CSV/PDF" button • Format data • Test export compatibility	Data Analyst	5	5	5	5	5	5	5	5	5	5	5

4.4 Sprint 4 Backlog - Alerts & Payment Integration

User Story	Tasks / Sub-tasks	Responsible	Effort	Days Remaining						
User Story 10	• Create alert form (title + message) • Store and push to users • Display in-app notifications • Test delivery	Backend Dev	13	12	10	9	9	9	9	9
User Story 14	• Design payment screen • Simulate D17/eDinar API • Store payment confirmations • Test transaction flow	Backend Dev	13	13	13	13	13	13	13	13

5 Sprint Retrospective & Performance

5.1 Sprint Satisfaction Summary

Sprint	Satisfaction Level	Explanation	Element NOT included
Sprint 1	100%	All core features delivered + polished UI	• All done
Sprint 2	90%	Route planner + GIS layers + advanced UX delivered	• See station's real-time status (occupied/free)
Sprint 3	70%	Delivered analytical layers but no review system	• Export data (Not needed in a front-end map) • Reviews (Not implemented)
Sprint 4	30%	Integrate advanced extensions in progress	• Admin Alerts • Payment/Booking system

5.2 Sprint Retrospective Analysis

5.2.1 What Went Well

- Strong collaboration: tasks were well distributed
- Good technical quality: UI polished, map responsive, clean code
- GeoJSON layers worked perfectly
- Leaflet.js integrations were smooth (heatmap, coverage, routing)
- Problem-solving improved over time (debugging JSON, fixing interactions)

5.2.2 What Could Be Improved

- Authentication simulated (LocalStorage, not database)
- Route planner could be enhanced with multiple stations
- Some UI panels could be modularized for scalability
- Dataset could include: charger power curves, real-time data
- Lack of thorough mobile testing

5.2.3 Improvements For Next Sprint

- Use Firebase or a backend API for real authentication
- Add server for storing reviews, user data, bookmarks
- Implement real-time station availability
- Improve performance with clustering for many stations
- Add CI/CD or deployment version on GitHub Pages

5.3 Burndown Charts

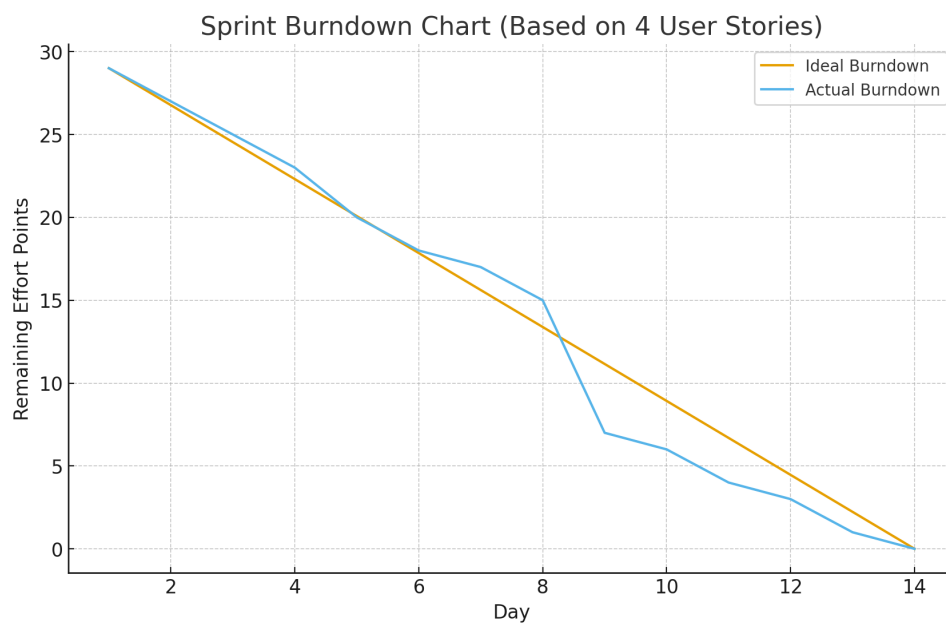


Figure 5.1: Sprint 1 Burndown Chart

All desired featured are met and the work is completely done.

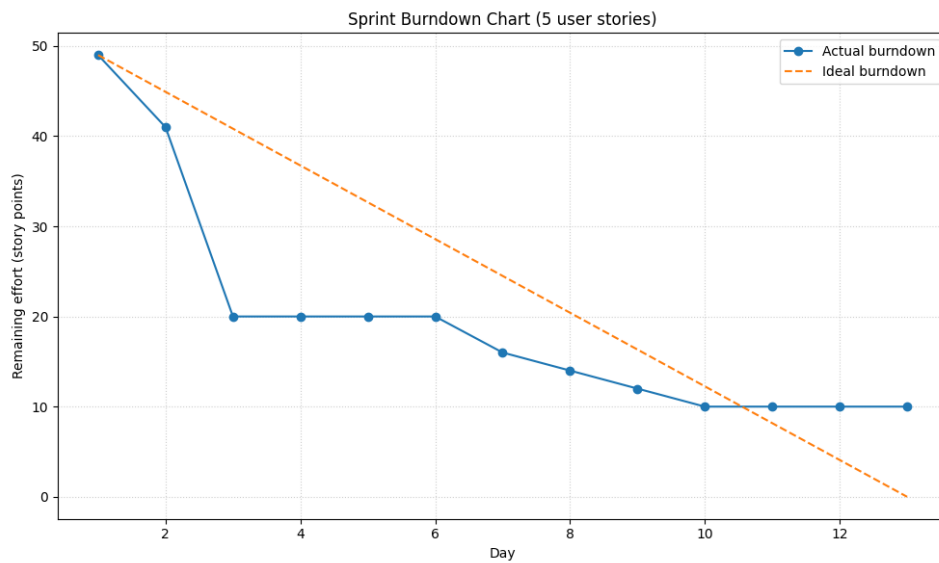


Figure 5.2: Sprint 2 Burndown Chart

Actual burndown (solid line): The total remaining effort each day across all 5 stories (in “effort units,” often story points or hours). Ideal burndown (dashed line): Even progress from start to zero by the sprint end. scope is not burning down fast enough

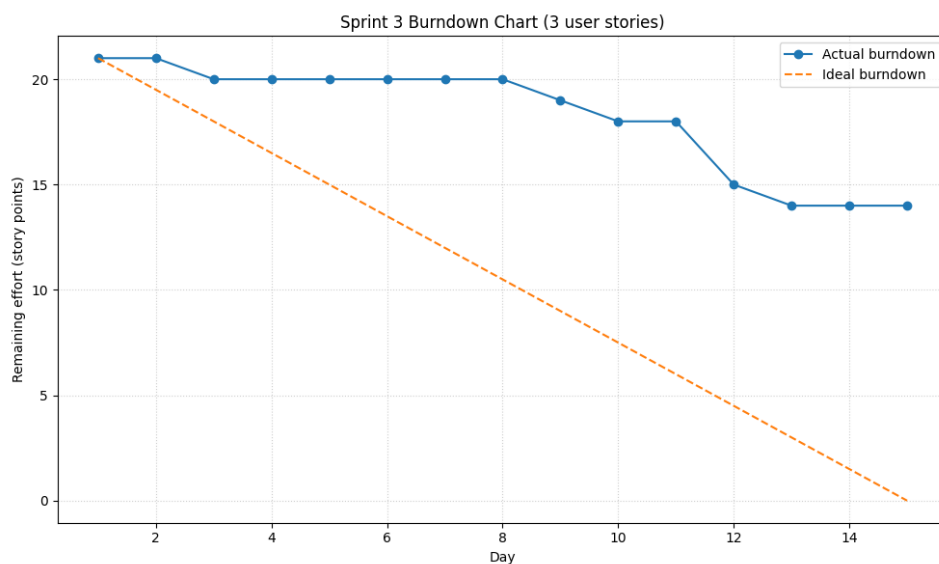


Figure 5.3: Sprint 3 Burndown Chart

Plateau behavior: Story 1 never burns down; Stories 2 and 3 plateau at 2 and 4 respectively. Risk indicator: Because the actual line remains well above the ideal, this sprint likely faced blockers, underestimation, or scope creep—common signals when work isn’t completing (no zeros).

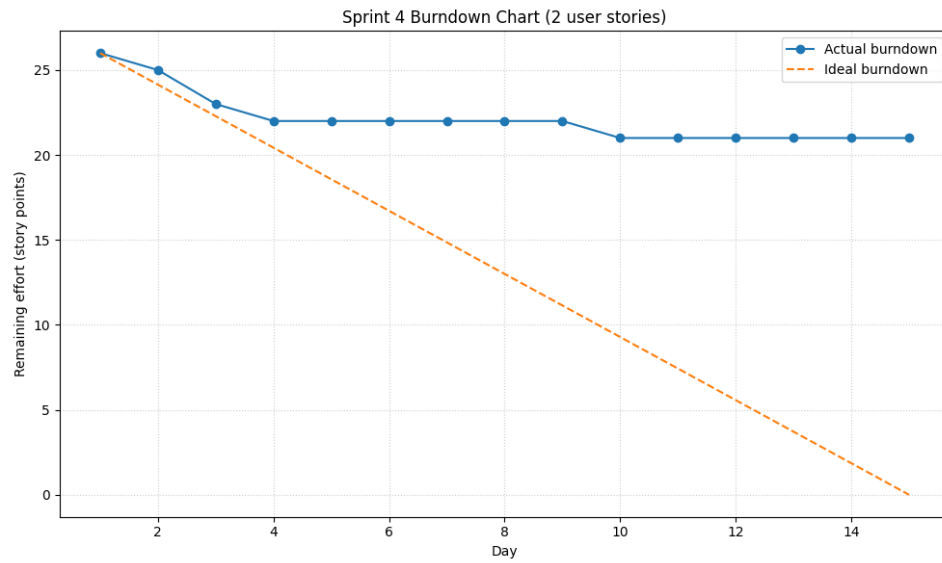


Figure 5.4: Sprint 4 Burndown Chart

The persistent flat line for Story 2 suggests a blocker or work not started. Story 1 makes early progress but stalls at 8. If completion is your target, you typically expect zeros at the end of the sprint for completed stories; none appear here.

6 Technical Infrastructure

6.1 Resources

- **Financial:** Supported by seed grants from green innovation programs ("Startup Tunisia," EU Green Deal)
- **Natural/Energetic:** Renewable-powered data servers (AWS North Africa Green Zone)
- **Technical:** GIS stack (PostGIS, Leaflet, ArcGIS), navigation APIs, IoT integration
- **Human Skills:** 3 GIS developers, 2 backend engineers, 2 mobile developers, 1 data scientist, 1 marketing & sustainability officer

6.2 Costs & Investments

6.2.1 Fixed Costs:

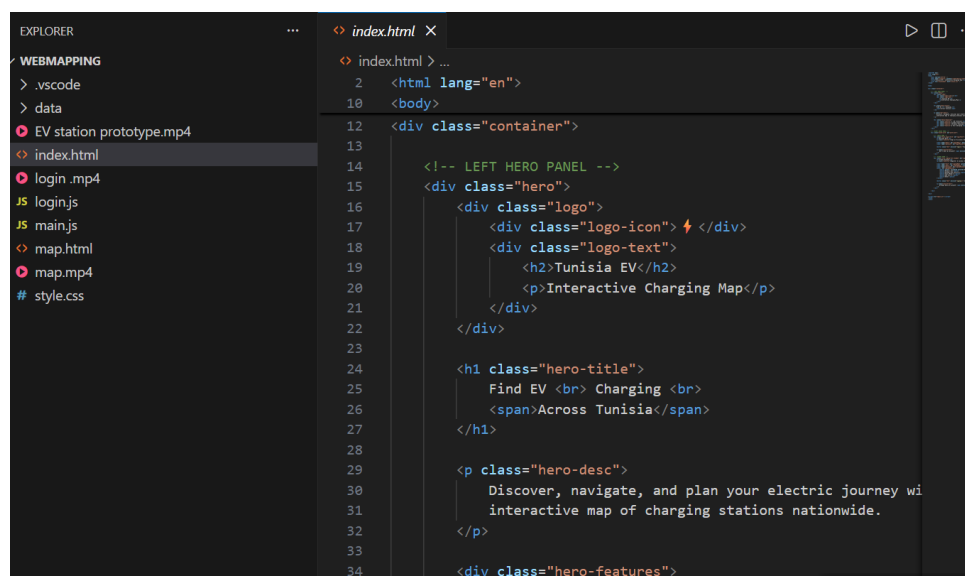
- App & platform development (40,000 TND)
- Server hosting & map API licensing (5,000 TND/year)
- Marketing and branding (3,000 TND)

6.2.2 Variable Costs:

- Maintenance & data updates (1,000 TND/month)
- Staff & operations (4,000 TND/month)
- Partnership fees / open data access

6.3 Platform Architecture

- Frontend: HTML, CSS, JavaScript with Leaflet.js
- Backend: Node.js/Express (planned)
- Database: PostgreSQL with PostGIS extension
- Mapping: OpenStreetMap, Google Maps API integration
- Hosting: AWS with green energy commitment

A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a file tree for a project named 'WEBMAPPING'. The main editor area displays the 'index.html' file, which contains HTML code for a web page. The code includes a container div, a hero panel with a logo, a hero title, a hero description, and a hero features section. The code is syntax-highlighted and includes line numbers from 2 to 34. The Explorer sidebar lists files: .vscode, data, EV station prototype.mp4, index.html, login.mp4, login.js, main.js, map.html, map.mp4, and style.css.

```
EXPLORER
WEBMAPPING
  .vscode
  data
  EV station prototype.mp4
  index.html
  login.mp4
  login.js
  main.js
  map.html
  map.mp4
  style.css

index.html
2  <html lang="en">
10 <body>
12 <div class="container">
13
14   <!-- LEFT HERO PANEL -->
15   <div class="hero">
16     <div class="logo">
17       <div class="logo-icon"> ⚡ </div>
18       <div class="logo-text">
19         <h2>Tunisia EV</h2>
20         <p>Interactive Charging Map</p>
21       </div>
22     </div>
23
24     <h1 class="hero-title">
25       Find EV <br> Charging <br>
26       <span>Across Tunisia</span>
27     </h1>
28
29     <p class="hero-desc">
30       Discover, navigate, and plan your electric journey wi
31       interactive map of charging stations nationwide.
32     </p>
33
34     <div class="hero-features">
```

Figure 6.1: VS Code Code Graph



Figure 6.2: Our Platform Interface Figure 1.0

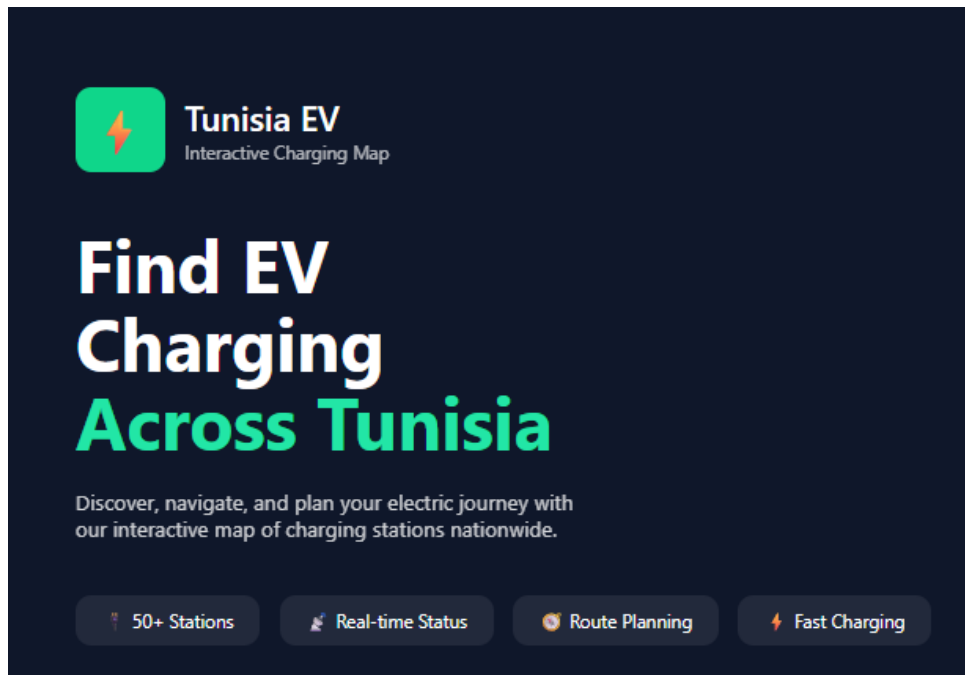


Figure 6.3: Our Platform Interface Figure 1.1

Create Account

Register to access the EV charging map

The form consists of four input fields stacked vertically, each with a light blue background and rounded corners. The first field contains the text 'sahar'. The second field contains the email address 'fekisahar0@gmail.com'. The third field contains three dots '...', indicating a password field. The fourth field contains the text 'Tesla Model 3' and a small downward-pointing chevron icon on the right side, indicating a dropdown menu. Below these fields is a large, solid green button with the text 'Create Account →' in white.

Already have an account? [Login](#)

Figure 6.4: The Sign Up Process Figure

Welcome Back

Sign in to access the EV charging map

fekisahar0@gmail.com

...



Sign In →

Don't have an account? [Sign up](#)

Figure 6.5: The Log In Process Figure

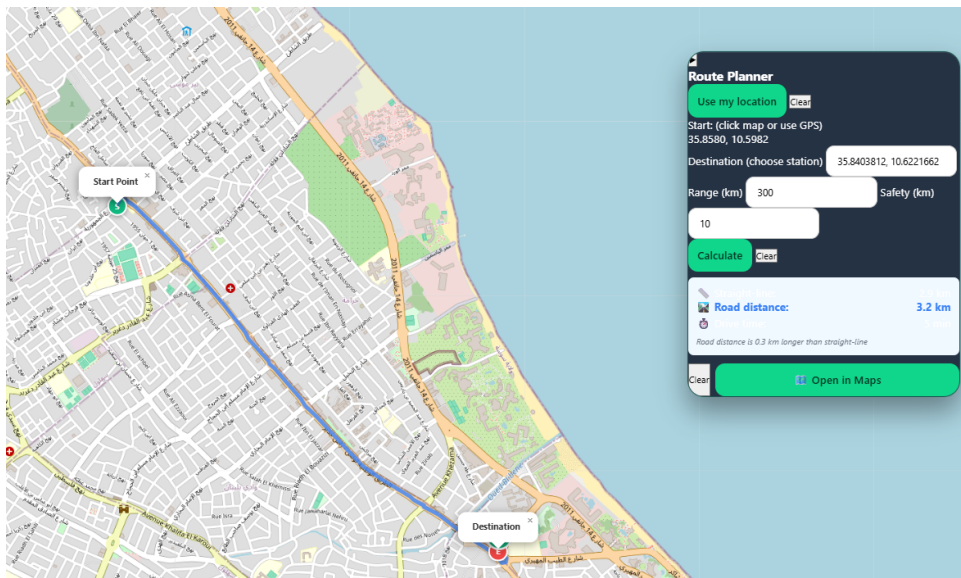


Figure 6.6: Map Road from a selected starting point to the destination

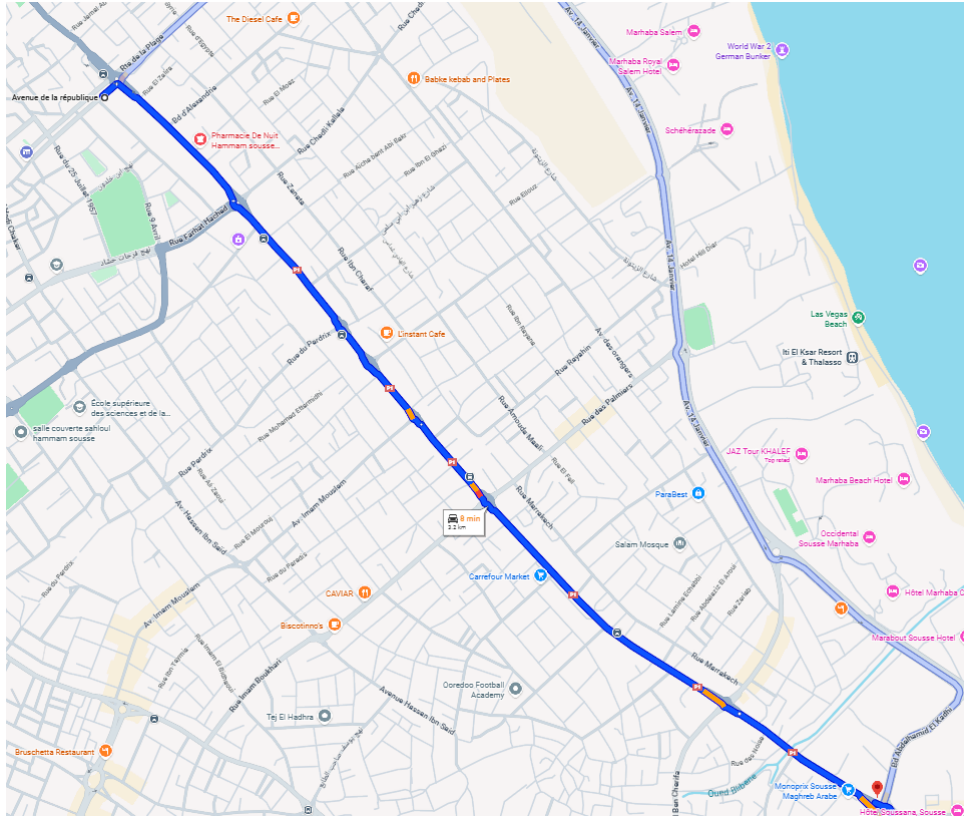


Figure 6.7: Map Road on Google Map to track real time

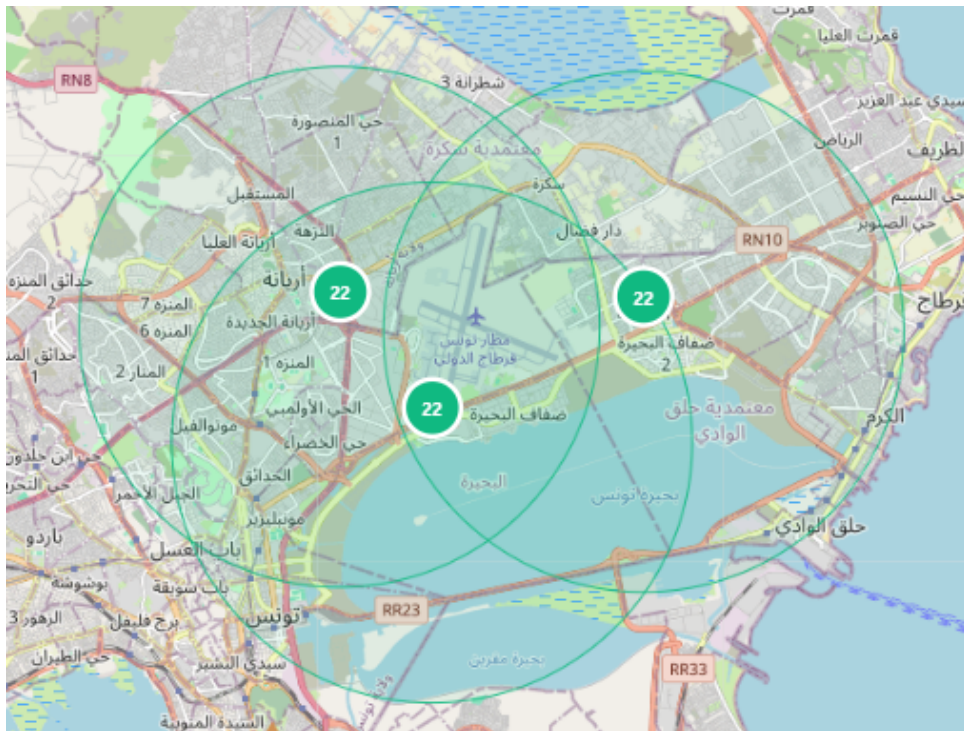


Figure 6.8: 5 KM Coverage of the Starting Point (localization of the user) and all the Charging Stations

Tunisia EV
Interactive Charging Map

sahar
Logout

Search by name / operatc

☐ 5 km coverage
 ☐ Density heatmap
 ☐ Unserved zones

Stations
15 shown • 15 total

TotalEnergies - Bizerte RN11	22 kW
TotalEnergies • Type 2	View
TotalEnergies - Jardins du Lac	22 kW
TotalEnergies • Type 2	View
TotalEnergies - Grombalia A1	50 kW
TotalEnergies • Type 2 / DC	View
TotalEnergies - Borjine A1	50 kW
TotalEnergies • Type 2 / DC	View
TotalEnergies - El Jem A1	50 kW
TotalEnergies • Type 2 / DC	View

Figure 6.9: Listing All the Charging Stations

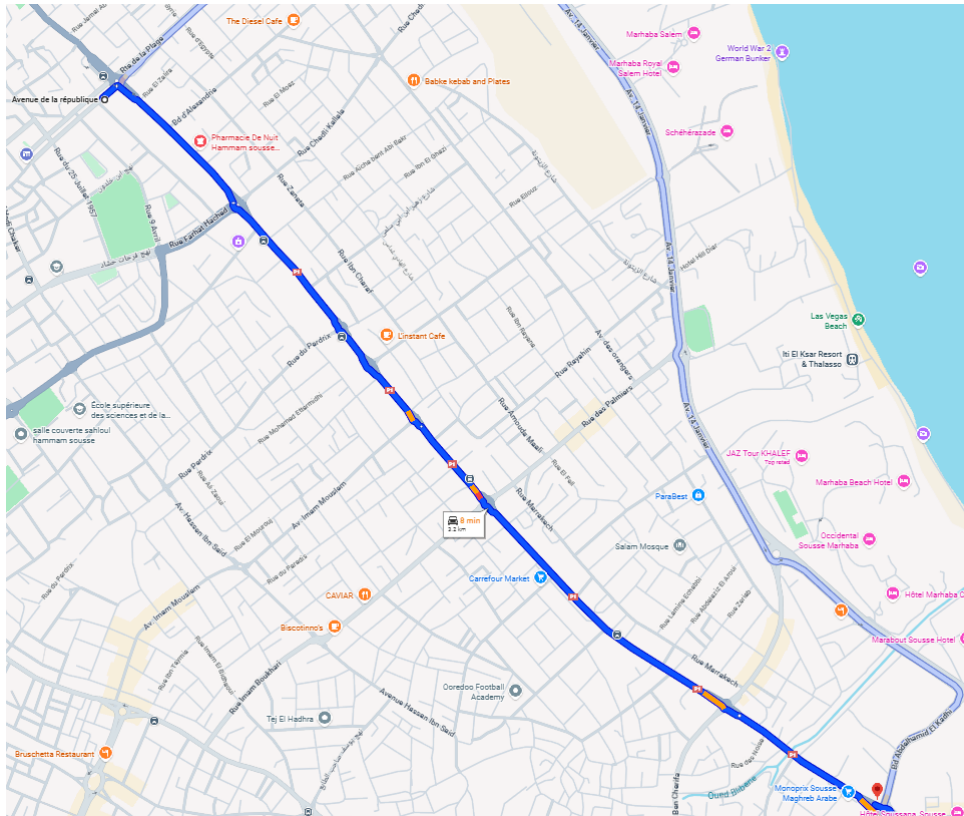


Figure 6.10: Map Road on Google Map to track real time

7 JIRA Software for Agile Project Management

7.1 Introduction to JIRA Software

JIRA Software by Atlassian is a powerful project management tool designed specifically for agile development teams. Throughout the SmartCharge TN project, JIRA served as our central hub for tracking progress, managing tasks, and maintaining transparency across all team members. By implementing JIRA, we established a professional workflow that enhanced collaboration, improved estimation accuracy, and provided real-time visibility into project status.

The platform's flexibility allowed us to customize workflows to match our Scrum methodology, with features including:

- **Product Backlog Management:** Centralized repository for all user stories and requirements
- **Sprint Planning:** Easy creation and management of sprint backlogs
- **Task Boards:** Visual Kanban-style boards for tracking work progress
- **Burndown Charts:** Automatic generation of sprint progress visualizations
- **Time Tracking:** Effort estimation and actual time spent monitoring
- **Reporting:** Comprehensive analytics on team velocity and sprint performance

7.2 How JIRA Enhanced Our Project Organization

Implementing JIRA brought significant improvements to our project management process:

7.2.1 Centralized Project Visibility

JIRA provided a single source of truth for all project artifacts. The dashboard (Figure 7.1) gave immediate visibility into project health, showing key metrics like sprint progress, remaining work, and team velocity.

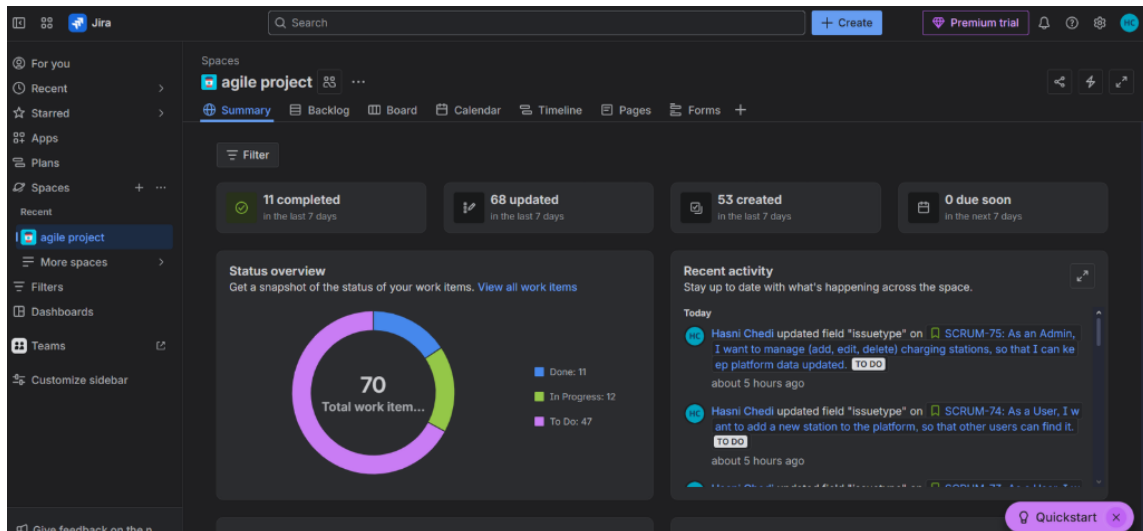


Figure 7.1: JIRA Project Summary Dashboard showing overall project status, active sprints, and key metrics

7.2.2 Structured Product Backlog Management

The product backlog (Figure 7.2) allowed us to maintain all user stories in a prioritized list, with clear acceptance criteria, estimates, and status tracking. This structured approach ensured that requirements were never lost and priorities were always clear.

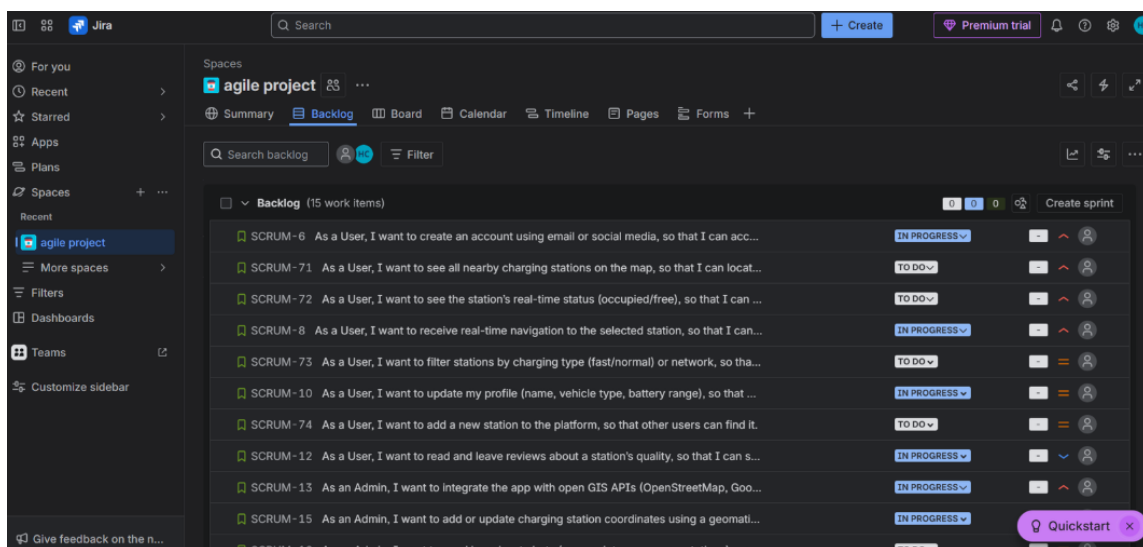


Figure 7.2: JIRA Product Backlog showing prioritized user stories with status, priority, and estimates

7.2.3 Visual Task Management

The Scrum board (Figure 7.3) transformed our workflow by providing a visual representation of work items moving through different stages (To Do, In Progress, Done). This visual management helped identify bottlenecks early and kept the team focused on sprint goals.

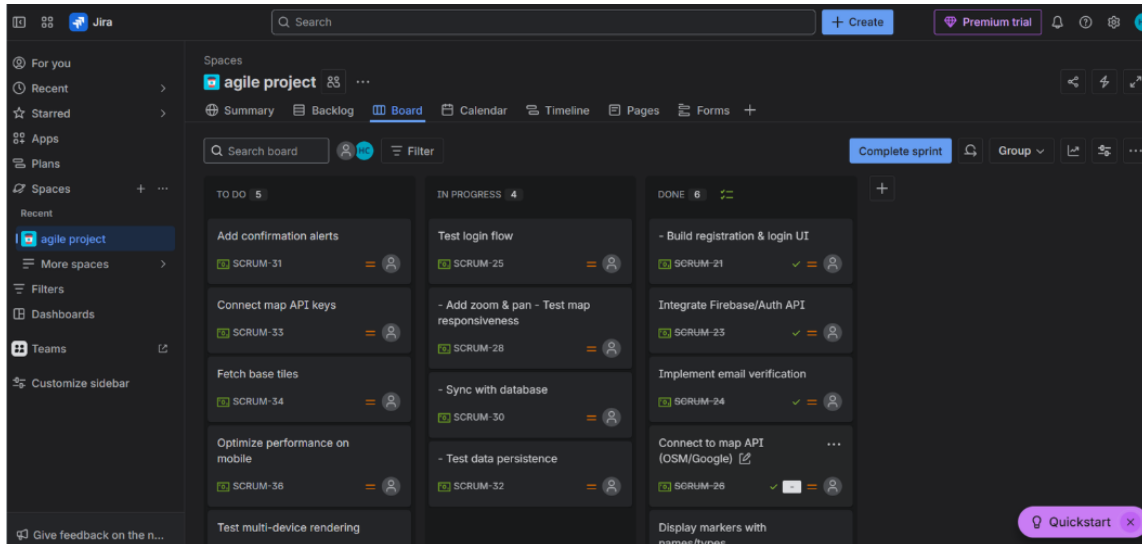


Figure 7.3: JIRA Task Board (Scrum Board) showing work items in different workflow states

7.2.4 Sprint-Specific Management

JIRA allowed us to create dedicated views for each sprint (Figures 7.4, 7.5, 7.6), containing only the user stories and tasks relevant to that sprint. This focused approach prevented scope creep and maintained sprint integrity.

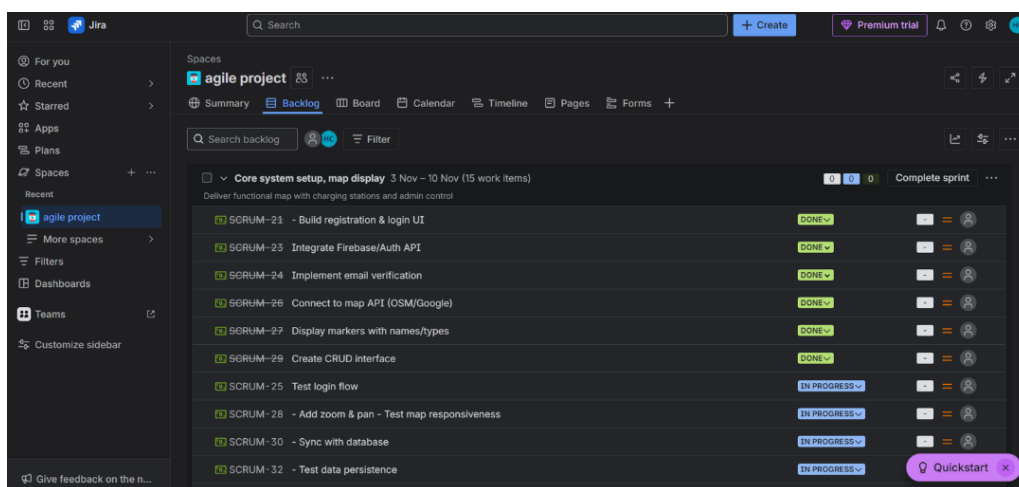


Figure 7.4: Sprint 1 in JIRA: Core system setup with detailed tasks and assignments

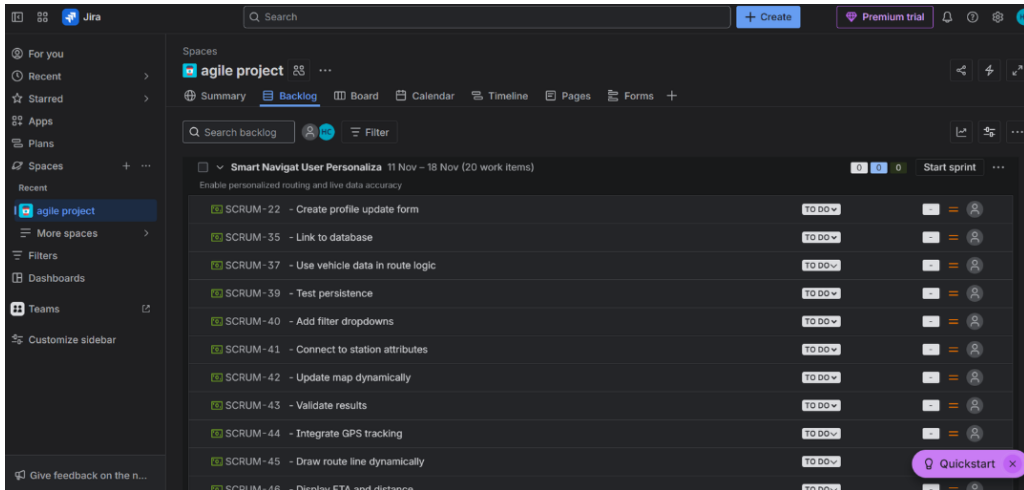


Figure 7.5: Sprint 2 in JIRA: Smart navigation features with progress tracking

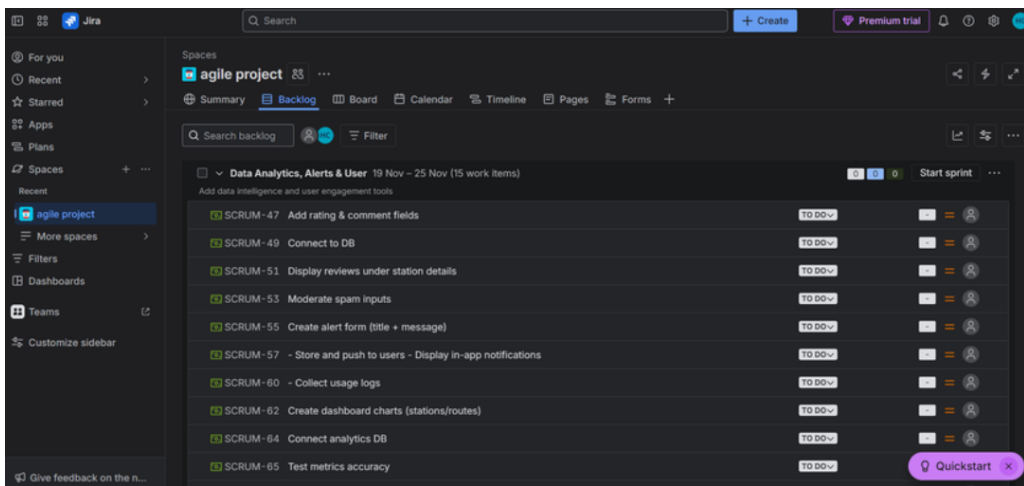


Figure 7.6: Sprint 3 in JIRA: Data analytics and user engagement features

7.3 Key Benefits Realized

The implementation of JIRA Software yielded several tangible benefits for the SmartCharge TN project:

7.3.1 Improved Team Coordination

JIRA's notification system and activity stream kept all team members informed about task updates, comments, and status changes. This real-time communication reduced misunderstandings and ensured everyone was aligned with project goals.

7.3.2 Enhanced Estimation Accuracy

By using JIRA's planning poker integration and historical data tracking, our team improved estimation accuracy over successive sprints. The ability to compare estimated vs. actual effort helped refine our planning process.

7.3.3 Better Stakeholder Transparency

The customizable dashboards and reports in JIRA allowed us to provide stakeholders with up-to-date progress information without manual status reporting. This transparency built trust and facilitated better decision-making.

7.3.4 Continuous Process Improvement

JIRA's retrospective tools and velocity tracking enabled data-driven discussions about process improvements. By analyzing sprint performance metrics, we could identify areas for improvement and implement changes systematically.

7.4 JIRA Implementation Strategy

Our 12-week Scrum transformation roadmap (as shown in the presentation) guided our JIRA implementation:

- **Weeks 1-2 (Foundation Setup):** Created JIRA project, defined team roles, established workflows
- **Weeks 3-4 (First Sprint Execution):** Conducted Sprint 1 entirely within JIRA with daily standups
- **Weeks 5-6 (Process Refinement):** Optimized JIRA workflows based on team feedback
- **Weeks 7-8 (Velocity Stabilization):** Used JIRA metrics to measure and stabilize team velocity
- **Weeks 9-10 (Advanced Practices):** Implemented story mapping and release planning in JIRA
- **Weeks 11-12 (Scaling Preparation):** Prepared JIRA for multi-team collaboration and final documentation

7.5 Lessons Learned and Best Practices

Our experience with JIRA revealed several best practices for agile project management:

- **Start Simple:** Begin with basic workflows and add complexity as the team becomes comfortable
- **Consistent Naming Conventions:** Establish clear naming conventions for epics, user stories, and tasks
- **Regular Maintenance:** Schedule weekly backlog grooming sessions to keep the product backlog organized
- **Training Investment:** Allocate time for team training on JIRA features to maximize adoption
- **Customization Balance:** Customize JIRA to match your workflow, but avoid over-customization that creates complexity

7.6 Conclusion on JIRA Usage

JIRA Software proved instrumental in the successful execution of the SmartCharge TN project. By providing a structured yet flexible framework for agile project management, JIRA enabled our team to:

- Maintain clear visibility into project progress and impediments
- Improve estimation and planning through historical data analysis
- Enhance collaboration through transparent task assignment and tracking
- Deliver consistent value through well-organized sprint cycles
- Adapt quickly to changing requirements while maintaining project focus

The integration of JIRA into our development process transformed how we managed the SmartCharge TN project, moving from ad-hoc task management to a professional, data-driven approach. The screenshots included in this chapter demonstrate how JIRA's various features were utilized to manage our product backlog, track sprint progress, and maintain overall project visibility.

For future projects, we recommend JIRA Software as an essential tool for any team practicing agile methodologies, particularly for complex projects requiring coordination between multiple roles and stakeholders.

8 Conclusion

SmartCharge TN successfully addresses Tunisia's EV charging infrastructure challenges through a real-time geospatial platform. By implementing Agile Scrum methodology, the team delivered incremental value through four sprints, resulting in a functional MVP that enables EV drivers to locate charging stations efficiently.

8.1 Key Achievements

- Developed Tunisia's first real-time, country-specific EV charging platform
- Implemented interactive mapping with real-time status updates
- Created route optimization for efficient EV travel
- Established a sustainable business model supporting Tunisia's green transition
- Successfully applied Scrum methodology with clear sprint planning and retrospectives

8.2 Future Vision

SmartCharge TN aims to become the leading smart mobility platform in North Africa by 2030. Future developments include:

- Integration with Tunisia's national digital mobility network
- Regional expansion through strategic partnerships
- Advanced analytics for energy operators and municipalities
- Mobile app development for iOS and Android
- IoT sensor integration for real-time station monitoring

8.3 Final Impact

The project directly supports Tunisia's 2030 National Energy Transition Strategy by:

- Reducing CO emissions through facilitated EV adoption
- Building public confidence in clean energy infrastructure
- Stimulating private investment in charging stations
- Creating green jobs in mobility, GIS, and IoT sectors
- Providing data-driven insights for urban planning

SmartCharge TN represents a significant step toward sustainable mobility in Tunisia, demonstrating how technology and agile project management can drive environmental and economic progress.